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*Environmental Spheres: How Planet Earth Works*

Over 13 billion years ago, the Big Bang occurred. This catalyst did not immediately create Earth, but allowed for a chain of unique processes that allowed for its formation 4.54 billion years ago. At this point, Earth was still just a clump of gas and dust, but four unique environmental spheres were created that allowed for life to exist on the planet. These spheres have not only been the genesis for Earth, but the moderator of how the planet has developed.

Earth's geography is affected by all four of its environmental spheres, the lithosphere, hydrosphere, biosphere, and atmosphere in unique ways. These environmental spheres have decided over the course of millions of years the way the Earth is shaped and changed, in both long periods of time and short periods of time. Each sphere has unique characteristics and processes that have changed the Earth's geography over time, interacting with each other and causing vast amounts of the Earth to look the way it does today. These environmental spheres all support unique environments in all parts of the world.

Firstly, the hydrosphere is the total amount of water on planet Earth, whether it be as big as the Pacific Ocean to what you carry around in your backpack. In geography, what the hydrosphere is studied on is the different landforms that have been made by or are water. These landforms are clouds, oceans, glaciers, streams, and lakes, all of which are made up of water in its liquid, solid, and gas forms. Within this sphere, the unique

hydrologic cycle that examines condensation, transpiration, evaporation, and precipitation is examined. For example, water from the Pacific Ocean might be evaporated and transformed into a cloud. This cloud will move towards the shore, going up to snow in the Sierra Nevada Mountains. Snow will melt into water, stream down a mountain into rivers like the American River and either meet the ocean or be used as usable non-salt water. At this point humans can drink the water, transforming it into urine or sweat, or using it to water plants and trees, which use transpiration to bring water back into the atmosphere. When this hydrologic cycle is in action, the atmosphere is used to transport the water through the air. When water evaporates, it becomes hydrogen and oxygen, making up a large portion of the atmosphere. This mixture makes up clouds and the concentration of it in the air is one of the biggest differences between dry and wet climates. The biosphere relies on the hydrosphere for survival, with all plants and animals requiring the resource to survive. Clouds and the different degrees of wetness and moisture are the difference between some plants and animals adapting drought tolerant traits and hydrophilic traits. Lithosphere is affected by the hydrosphere through erosion, weathering, and sedimentation. Through streams and events that introduce moisture into rocks and breaking them down the hydrosphere is one of the largest factors in shaping landforms. In all, the hydrosphere is one of the most integral parts to sustain life on Earth as well as a geographical function on this planet.

Next, the lithosphere consists of the mantle and crust, all rocks that are on Earth. These rocks are as large as tectonic plates to small grains of sand on a beach. Ultra large rock floating on the mantle is known as tectonic plates which undergo processes that affect large parts of the world with each movement. For instance, at points where these tectonic plates meet are called fault lines. These fault lines are where these plates come

together, pull apart, or slide past each other. When these events happen, they cause earthquakes, seafloor spreading, and cause mountain ranges to rise. For example, when an oceanic plate is pushed under a continental plate, it is melted under the mantle and creates mountains on the continental plates. These mountains are then shaped by the hydrosphere weathering and eroding them down, creating unique landforms in different parts of the world. Depending on the amount of moisture, deserts will often have rocks that are affected by water a lot, or forests will often have rocks that are able to withstand a lot of water. The biosphere lives on the lithosphere, interacting with the large mountains and caves as a place to live in. Animals and plants use the rocks created to live and grow depending on what types of landforms were created. A goat living on the mountain can climb steep ledges whereas a dog from a prairie could be able to go far distances over flat land. Wind from the atmosphere is often blocked by the mountain ranges created, being the difference between an area receiving water and not receiving it. The lithosphere is the large part of the Earth that makes up what our planet is shaped in a geological fashion, and works with the other spheres to sustain life on Earth.

Therefore, the atmosphere is the layer of gases that are above the crust held on the Earth by gravity. This atmosphere is what humans, animals, and plants take in every day. The atmosphere is made up of multiple different layers, however the troposphere is the air we breathe and where all weather takes place. In this sphere, processes like the carbon cycle, wind, and convection are what shape the environment. The carbon cycle acts like the hydrologic cycle, carbon in the atmosphere is taken into plants through photosynthesis, which decay, become fossil fuels, and are released into the atmosphere again through carbon emissions. The amount of carbon in the atmosphere warms up the Earth through convection, the atmosphere keeping the radiation of the sun's heat on

Earth. Both of these processes are variables in the wind process, the act of air going from high pressure areas to low pressure areas. The way the wind blows has a very large part in shaping the landscape of an environment. For example, wind is what carries moisture and thus precipitation to certain areas. If there is a large mountain range in the way, the clouds will not get over and the leeward side will become very dry. Birds and insects use the wind to get to places, and you will see flocks of birds in a V formation to combat the wind. This in turn causes migration and other effects on the environment like seasonal periods where there are a lot of birds or not a lot. In all, the atmosphere is what makes up the air on Earth, what the wind moves to shape and take things places.

Finally, the biosphere is the total of all living things on Earth, humans, animals, and plants. Arguably, this is the most impactful force that affects the planet we live on currently. This is due to humans being able to manipulate the processes in all other spheres, not just the biosphere in the current day. Going back to thousands of years ago, when humans, animals, and plants were all pretty much equal, the most significant thing occurring in the biosphere was the food chain. This is the act of plants going through photosynthesis and getting the sun's energy. Then being consumed by herbivores, and then those animals, carnivores, consuming those animals. The Circle of Life, with decomposers taking the carcasses of the carnivores, recycles them back into the soil the plants can use. When disrupted, the environment can become very different from what it has been. Now that humans have become the de facto apex predator, they have a high amount of control over all spheres, especially the biosphere. However, it is notable that when they are not included or make changes, the biosphere still has one of the largest processes. For instance, when the wolves of Yellowstone were disrupted, a trophic cascade occurred that made the river flood and changed the landscape of the

national park. Furthermore, when the wolves were reintroduced, it allowed the park to return back to normalcy and reshaped the landscape of Yellowstone again. Since the biosphere is all living things on Earth, it is what differentiates the Earth from all other planets in the universe that we know of.

In conclusion, the four environmental spheres all work in collaboration to shape Earth to be the planet that we can live on. One last example of how these spheres work together is the place we live in, California. Sitting on the San Andreas fault, the Pacific Plate and North American plate had uplift in the lithosphere that created the Sierra Nevada mountain range. This mountain range, as well as the tectonic activity along the fault, has created vast amounts of sedimentary rocks that line beaches like Laguna Beach and Corona Del Mar. Where these beaches are, they meet the Pacific Ocean, which is subject to the westerly wind of the atmosphere pushing waves upon the shore for thousands of years. These waves of the hydrosphere eroded the beach, creating the perfect environment for wildlife to thrive. This allows for the biosphere: seagulls, crabs, and other marine life to prosper in this area in coves and bays. When looking at the planet, examples of these processes can be seen everywhere one looks. If one sphere did not have the processes it had, the planet would not have looked the way it does. These miracles are what allows Earth to be Earth, and why we've yet to find another planet that can sustain life like ours.